
The Standard Incomplete Markets (SIM) Model

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Individual household problem

The “standard incomplete markets” model (steady state)

- ❖ Individual household i optimizes

$$\max_{\{a_{it}, c_{it}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it})$$

subject to period-by-period budget constraint and borrowing constraint

$$a_{it} + c_{it} = (1 + r)a_{i,t-1} + Ze_{it} \quad a_{it} \geq \underline{a}$$

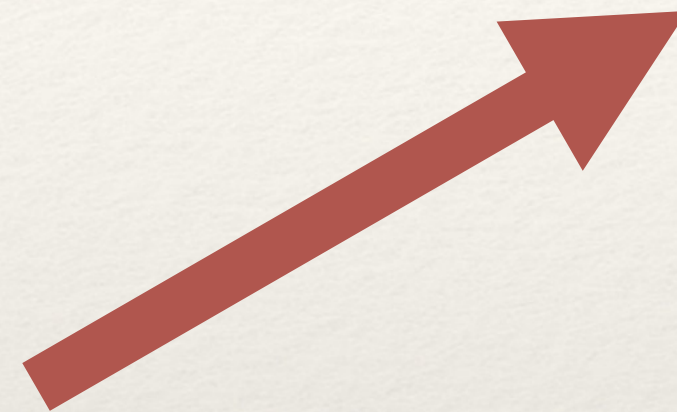
- ❖ Exogenous income state e_{it} follows Markov chain, which we'll usually normalize to 1, Z scales aggregate after-tax income
- ❖ Initial assets $a_{i,-1}$ taken as given, standard assumptions on u (CRRA)

Can convert sequential form to Bellman equation

$$\max_{\{a_{it}, c_{it}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it})$$

$$a_{it} + c_{it} = (1 + r)a_{i,t-1} + Ze_{it}$$

$$a_{it} \geq \underline{a}$$



$$V(e, a) = \max_{c, a'} u(c) + \beta \mathbb{E}[V(e', a') | e]$$

$$s.t. \ a' + c = (1 + r)a + Ze$$

$$a' \geq \underline{a}$$



Solved by **policies** $a'(e, a)$ and $c(e, a)$ in two state variables,
exogenous income state e and
endogenous asset state a

Solving the Bellman equation

- ❖ Policies $a'(e, a)$ and $c(e, a)$ satisfy standard **first-order condition**

$$u'(c) \geq \beta \mathbb{E}[V_a(e', a') | e]$$

with equality unless borrowing constraint binds, and **envelope condition**

$$V_a(e, a) = (1 + r)u'(c)$$

- ❖ Combined, same as sequential **Euler equation** $u'(c_{it}) \geq \beta(1 + r)\mathbb{E}_t[u'(c_{i,t+1})]$
- ❖ Can use first-order and envelope conditions to iterate backward on V_a and policies
 - ❖ Best way: interpolation and “endogenous gridpoints” (Carroll 2006)
 - ❖ Iterating until convergence gives V_a and steady-state policies on a grid

See computation supplement (on GitHub) for more!

```
def backward_iteration(Va, Pi, a_grid, y, r, beta, eis):
    # step 1: discounting and expectations
    Wa = (beta * Pi) @ Va

    # step 2: solving for asset policy using the first-order condition
    c_endog = Wa**(-eis)
    coh = y[:, np.newaxis] + (1+r)*a_grid

    a = np.empty_like(coh)
    for e in range(len(y)):
        a[e, :] = np.interp(coh[e, :], c_endog[e, :] + a_grid, a_grid)

    # step 3: enforcing the borrowing constraint and backing out consumption
    a = np.maximum(a, a_grid[0])
    c = coh - a

    # step 4: using the envelope condition to recover the derivative of the value function
    Va = (1+r) * c**(-1/eis)

    return Va, a, c
```

Basic backward iteration takes just 9 lines of standard Python code

Consolidated in `sim_steady_state.py`, GitHub links to supplementary notebook, video lectures, and also 2x sped-up version

Distribution of households

Solved household problem, now aggregate

- ❖ We've solved problem facing individual household
- ❖ Now aggregate into economies with a **continuum** of such households
 - ❖ Soon will put in **general equilibrium** ...
 - ❖ But for now interested in properties of “**partial equilibrium**” model, i.e. taking return r as given
- ❖ This is a **heterogeneous-agent economy**
 - ❖ Has a **distribution** of households across the two states, e and a

What is distribution of households?

- ❖ In principle, it's a **measure** μ
- ❖ If finitely many e , then can define $\mu(e, \mathbb{A})$ separately for each e , as measure on subsets $\mathbb{A}$ of the asset space

- ❖ Law of motion

$$\mu_{t+1}(e', \mathbb{A}) = \sum_e \mu_t(e, (a')^{-1}(e, \mathbb{A})) \cdot P(e, e')$$

where $P(e, e')$ is transition probability, $(a')^{-1}(e, \cdot)$ is inverse of policy $a'(e, \cdot)$

- ❖ Measure of $\mathbb{A}$ today is sum of measures yesterday that send you there today

Why measure?

- ❖ You might want some nice density function ...
 - ❖ But if $\beta(1 + r) < 1$, there will be a **positive mass at borrowing constraint**
 - ❖ (If $\beta(1 + r) \geq 1$, can show everyone's assets will diverge to ∞ , so we generally don't consider that case...)
 - ❖ If finitely many e , this implies **discrete distribution** with only mass points!
 - ❖ Countably many histories of e since last time hitting constraint.
- ❖ So for generality, we assume an arbitrary measure over assets
 - ❖ Will revisit much later when we build a “smoother” model

Calculating distribution in practice

```
def get_lottery(a, a_grid):  
    # step 1: find the i such that a' lies between gridpoints a_i and a_(i+1)  
    a_i = np.searchsorted(a_grid, a) - 1  
  
    # step 2: obtain lottery probabilities pi  
    a_pi = (a_grid[a_i+1] - a)/(a_grid[a_i+1] - a_grid[a_i])  
  
    return a_i, a_pi
```

Approximate distribution by point masses on finite grid; when asset policy $a'(e, a)$ lies between two gridpoints, convert it to “lottery” between gridpoints with same expectation

```
@numba.njit  
def forward_policy(D, a_i, a_pi):  
    Dend = np.zeros_like(D)  
    for e in range(a_i.shape[0]):  
        for a in range(a_i.shape[1]):  
            # send pi(e,a) of the mass to gridpoint i(e,a)  
            Dend[e, a_i[e,a]] += a_pi[e,a]*D[e,a]  
  
            # send 1-pi(e,a) of the mass to gridpoint i(e,a)+1  
            Dend[e, a_i[e,a]+1] += (1-a_pi[e,a])*D[e,a]  
  
    return Dend
```

Also in
sim_steady_state.py,
notebook, and videos.

Steady state of aggregate model

What is a steady state of model?

❖ Consists of:

❖ **policy functions** $a'(e, a)$ and $c'(e, a)$ that solve Bellman equation

❖ **distribution** $\mu(e, \mathbb{A})$ that satisfies steady-state law of motion

$$\mu(e', \mathbb{A}) = \sum_s \mu(e, (a')^{-1}(e, \mathbb{A})) \cdot P(e, e')$$

❖ Can show such a “stationary distribution” exists and is unique if $\beta(1 + r) < 1$

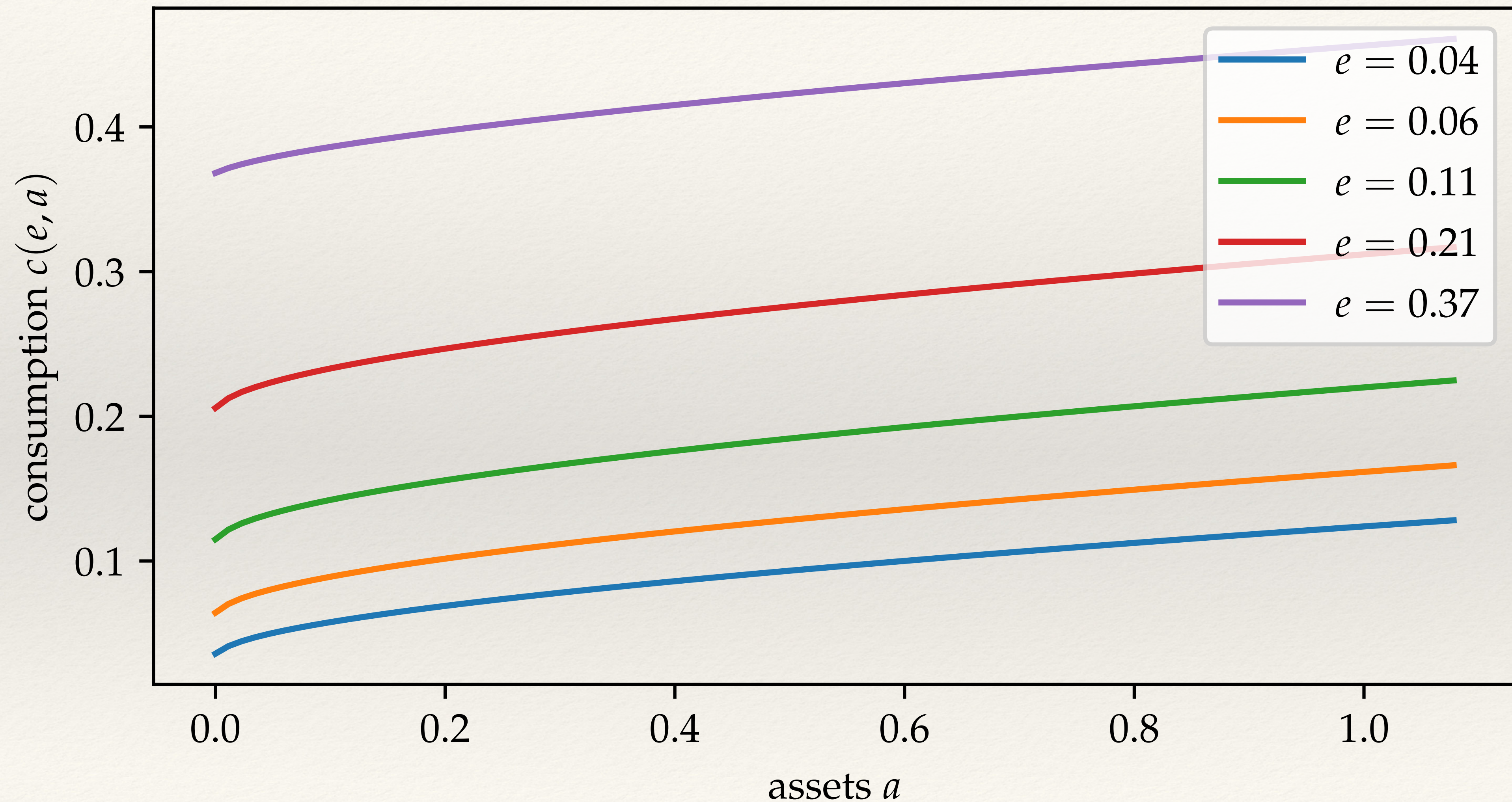
❖ **Aggregate assets and consumption:**

$$A = \int a d\mu = \int a'(e, a) d\mu \qquad C = \int c d\mu$$

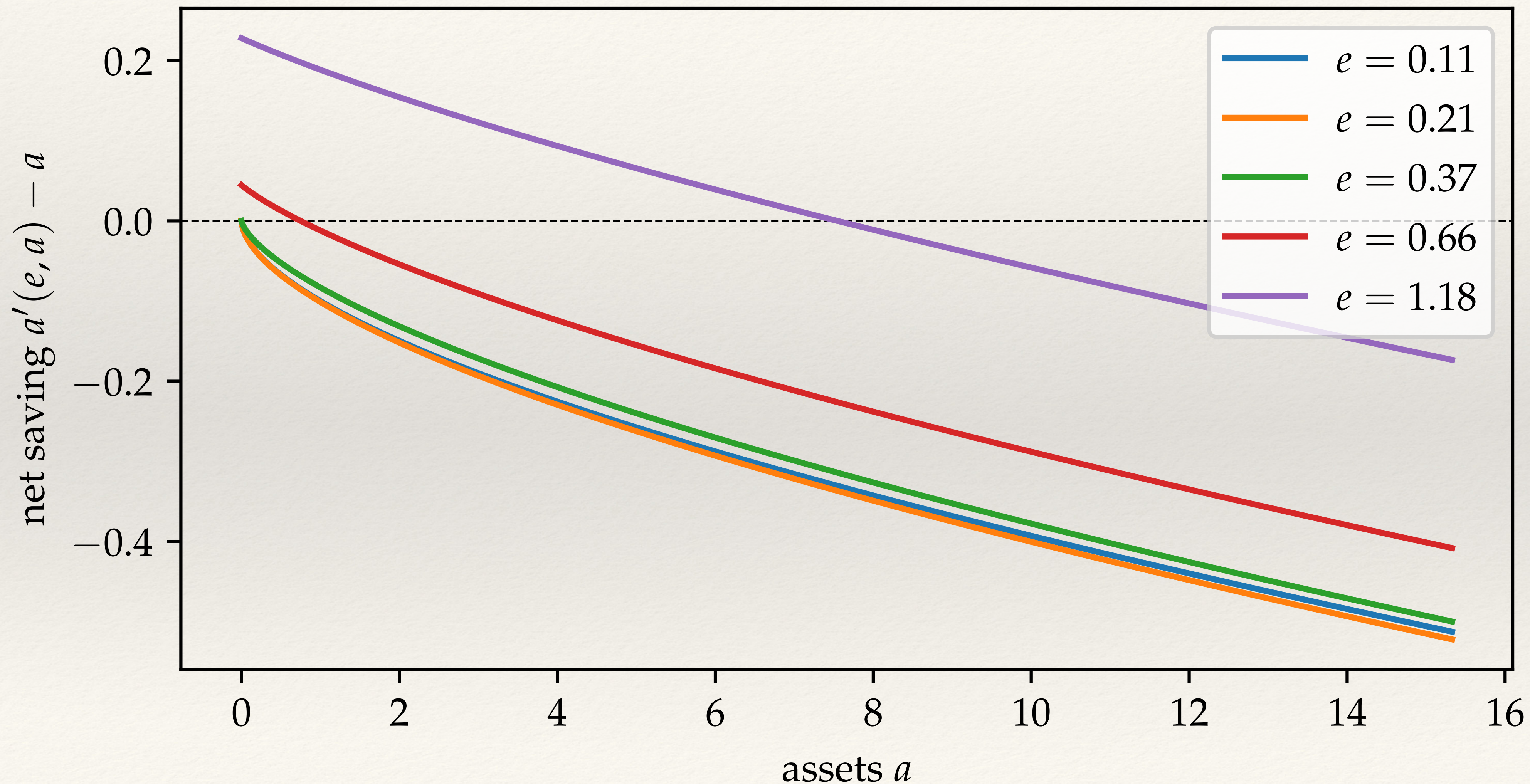
Nice features of model

- ❖ Captures key features of **consumption-saving problem** with risk
 - ❖ income smoothing, precautionary savings, etc.
- ❖ Endogenously generates **wealth distribution** (of assets a)
- ❖ Consistent with high **marginal propensities to consume (MPCs)** out of cash on hand, here $mpc(e, a) \equiv (\partial c(e, a) / \partial a) / (1 + r)$
- ❖ Unlike representative-agent model, steady-state **asset demand not infinitely elastic** in r , so r can be endogenous
- ❖ Easy to extend: other shocks, preference heterogeneity, endogenous labor, life-cycle structure, other assets...

Consumption functions: increasing, concave

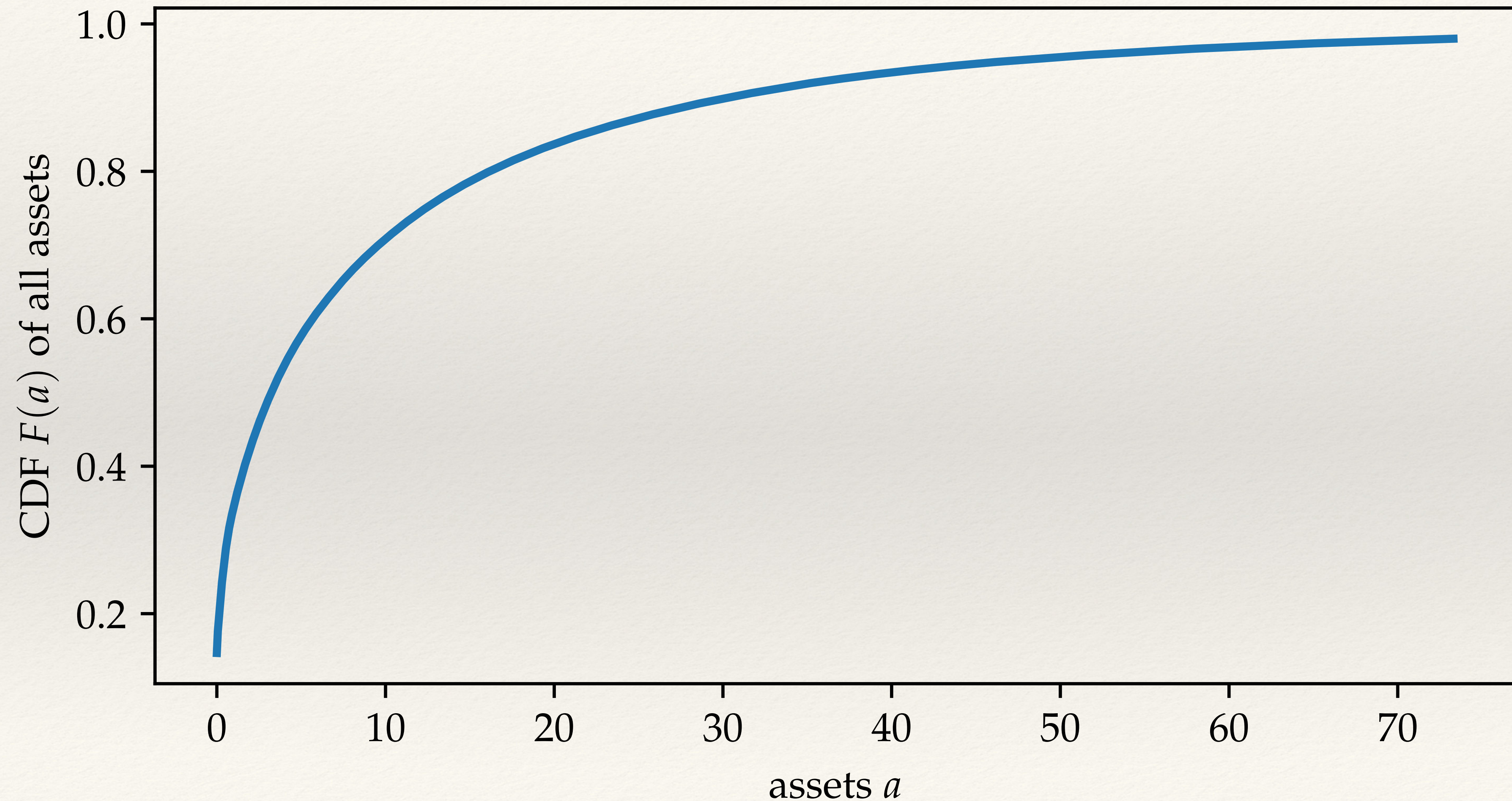


Buffer-stock behavior for each household



“Target” asset levels increase dramatically as we go to higher e , leading to inequality in stationary wealth distribution

Rich asset distribution, endogenous wealth inequality



Calibration of model

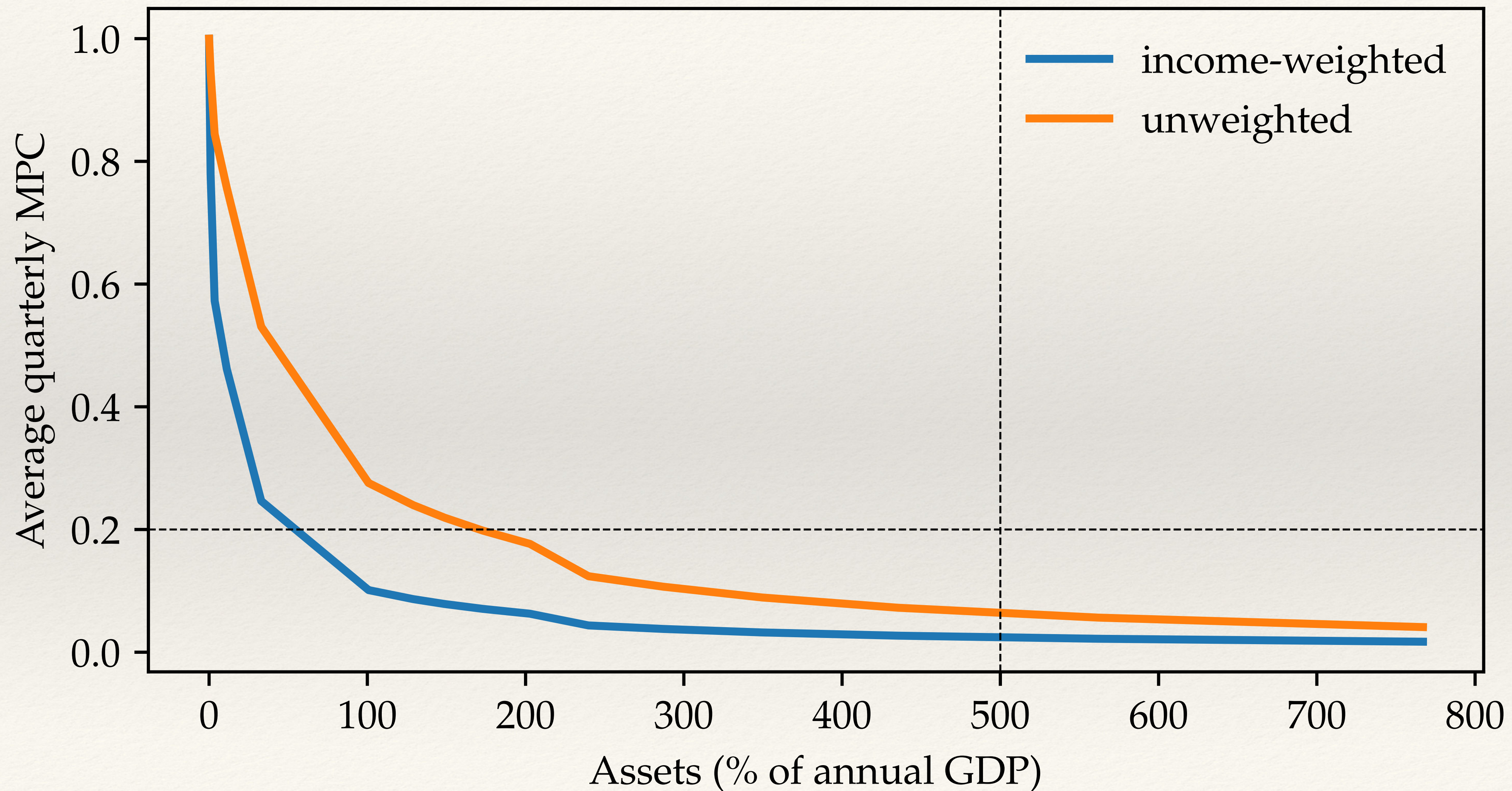
What parameters do we need to calibrate?

- ❖ Calibrate to quarterly frequency
- ❖ **Income process e :** [usually normalize average e to 1, entire ss scales in Z]
 - ❖ calibrate as discrete approximation of lognormal AR(1)
 - ❖ annual persistence $\rho = 0.91$, cross-sectional sd $\sigma = 0.92$ (IKC paper), rough approximation of pretax income process in US
 - ❖ 11-point Rouwenhorst approximation (see supplement for details)
- ❖ **Real rate r** to 2% annually, borrowing **constraint $\underline{a}$** to 0, **utility** to $u(c) = \log c$
- ❖ One parameter remains: **discount factor β**

Two common strategies for calibrating β

- ❖ Calibrate to hit target for **aggregate assets**, taken from data
 - ❖ Our Ann Rev calibration: assets A at 500% of GDP, given after-tax labor income Z of 70% of GDP, following US
 - ❖ (Some others target lower A , interpreted as some notion of “liquid” assets)
- ❖ Calibrate to average **marginal propensity to consume**, also taken from data
 - ❖ Our Ann Rev calibration: average income-weighted quarterly MPC of 0.2
- ❖ Problem: **tradeoff** between two, β that matches one fails other

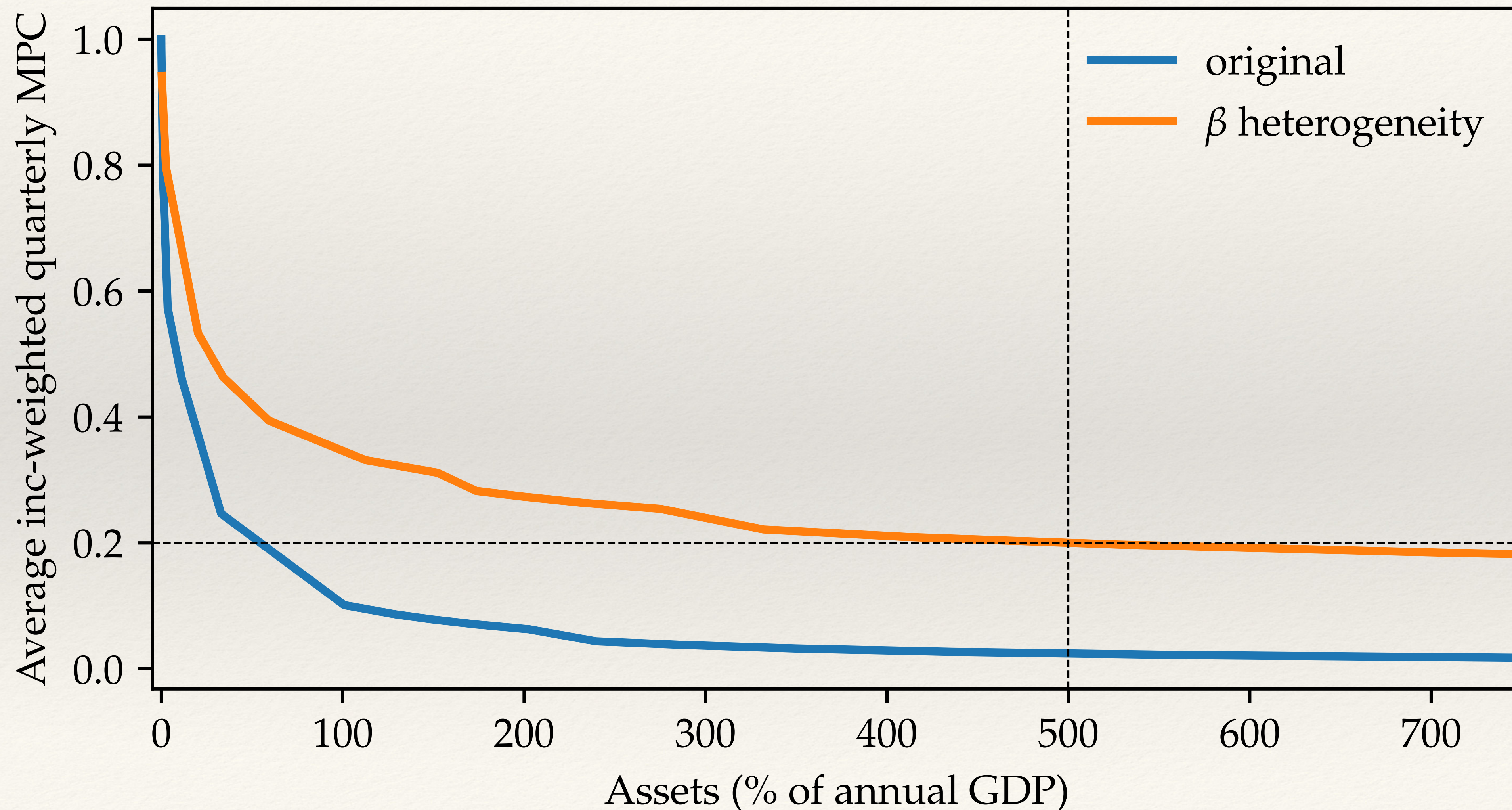
Asset-MPC tradeoff as we vary β



Our solution: β heterogeneity

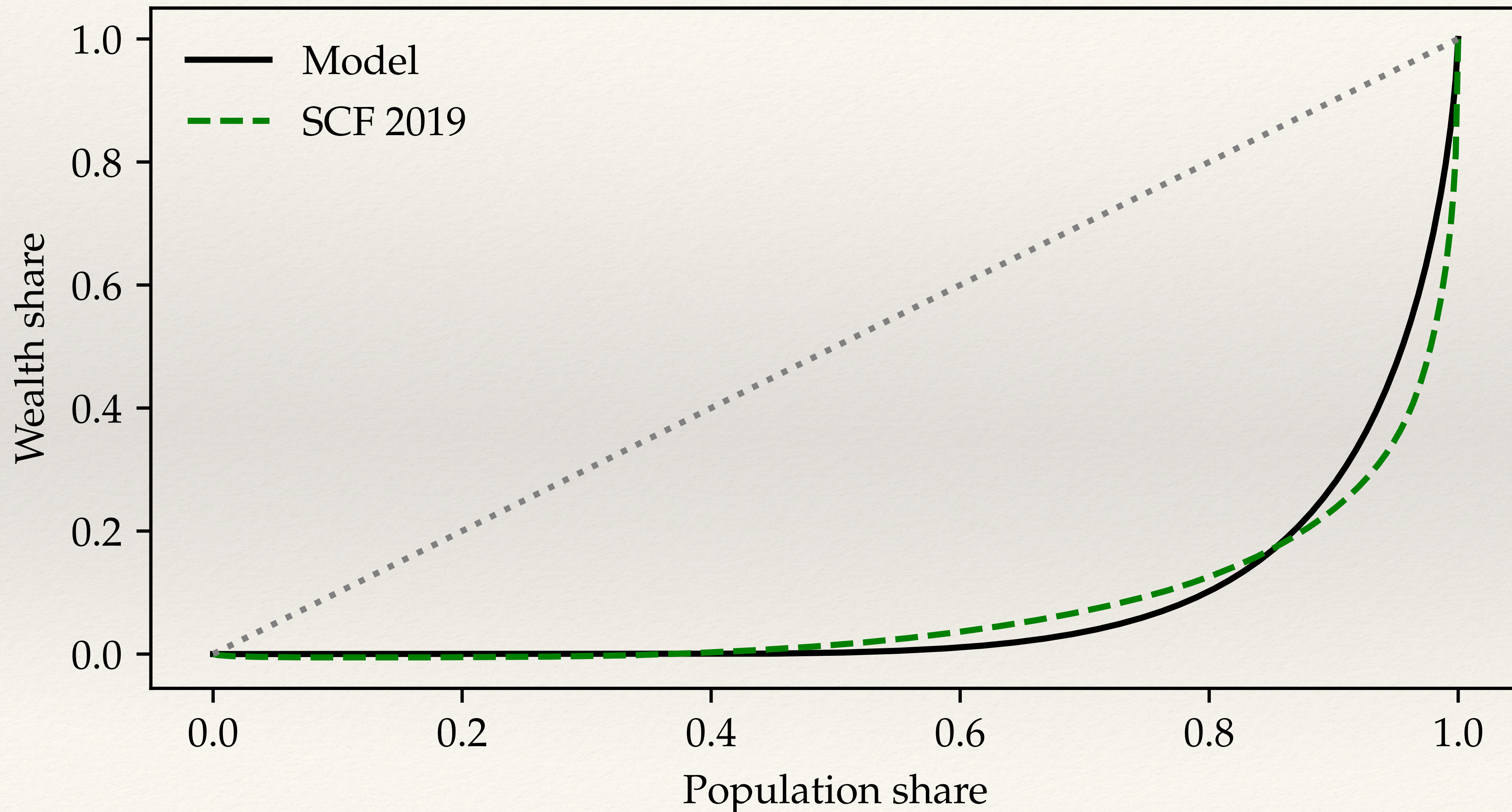
- ❖ By **mixing different β** , we can target both aggregate assets and MPCs
- ❖ Exogenous state is now (β, e) , can make β either permanent or stochastic
 - ❖ Stochastic version limits polarization into “spenders” vs. “savers”
- ❖ We’ll be inspired by Annual Review stochastic β process (independent of e):
 - ❖ Four equispaced β s with equal shares, each household gets fresh β draw 1% of time
 - ❖ Loosely interpret as new draw of preferences every “generation” (25 years)
 - ❖ Calibrate to **hit both asset (500% of GDP) and income-weighted MPC (0.2) targets**
 - ❖ Calibrated quarterly β s: approximately 0.94, 0.96, 0.98, 1.0
 - ❖ [For next few lectures, will also calculate “bonds only” calibration (assets = 100% of GDP).]

New vs. old asset-MPC tradeoff



(In β heterogeneity line, we keep gap between high and low β fixed as we vary the mean.)

Untargeted moment: Lorenz curve vs. US data



Model not bad at all since distribution is untargeted, but overstates “middle-class” wealth (50th to 90th percentiles), understates wealth in upper tail (difficult to match without other features)

Partial equilibrium dynamics

Time-varying aggregate inputs to household problem

- ❖ Revisit individual household problem [ignoring β process for notational simplicity]

$$\max_{\{a_{it}, c_{it}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it})$$

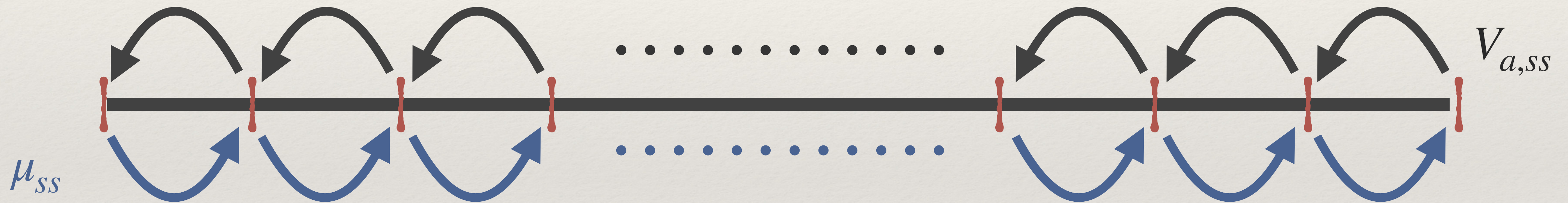
allowing returns r_t^p and aggregate after-tax income Z_t to vary over time

$$a_{it} + c_{it} = (1 + r_t^p) a_{i,t-1} + Z_t e_{it} \quad a_{it} \geq \underline{a}$$

- ❖ Add p to emphasize that r_t^p is **ex-post** return from $t - 1$ to t , determined at date t
- ❖ Assume distribution of $a_{i,-1}$ is steady state, perfect foresight over $\{r_t^p, Z_t\}_{t=0}^{\infty}$ from date 0 onward (“MIT shock”)

Solution uses similar iterations to steady state

1. Start with $V_{aT} = V_{a,ss}$ and iterate backward T times, using time-varying r_t^p and Z_t to obtain policies $a'_t(e, a)$, $c_t(e, a)$ at $t = 0, \dots, T - 1$



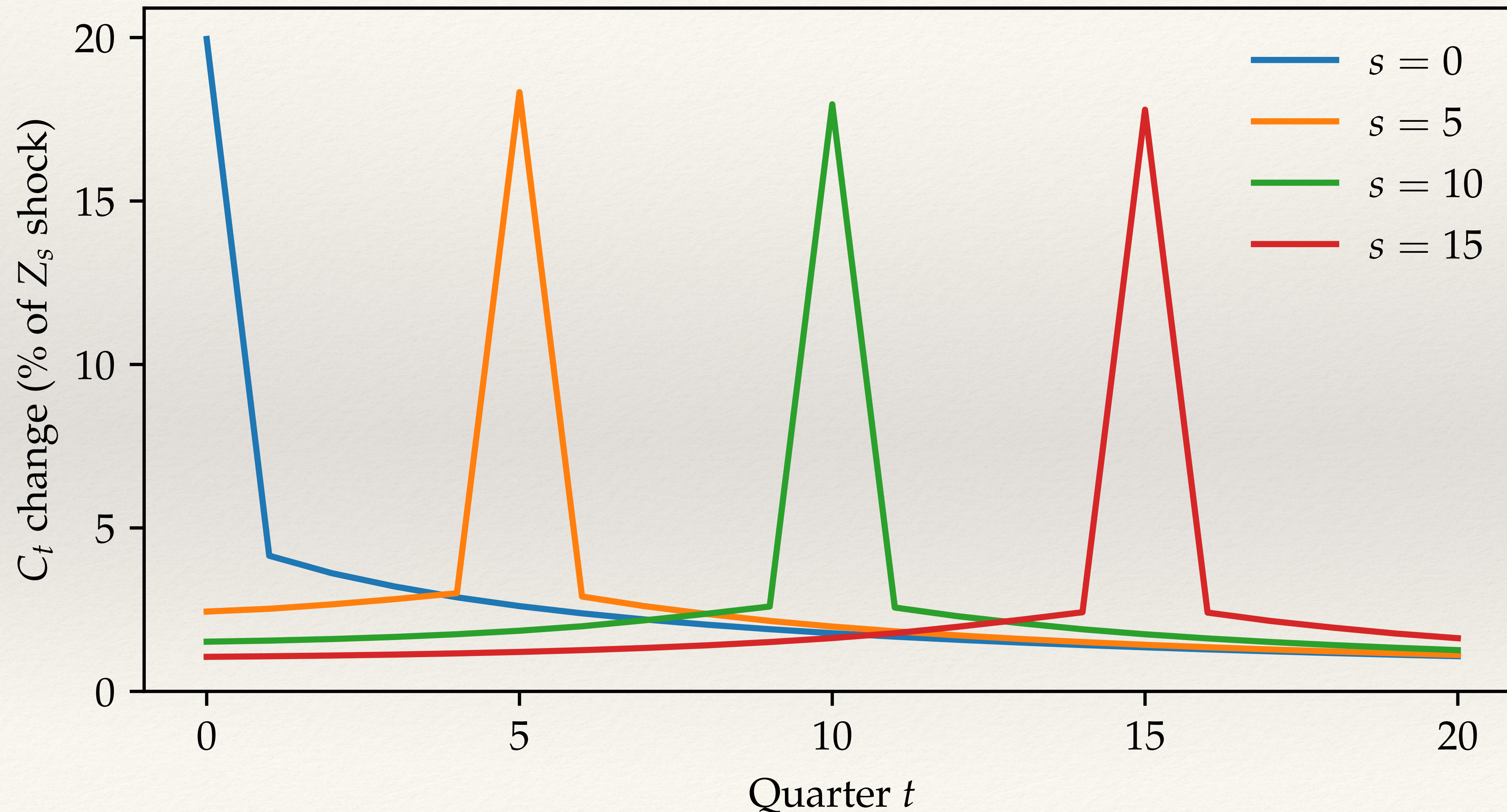
2. Start with distribution $\mu_0 = \mu_{ss}$ and iterate forward $T - 1$ times, using time-varying policy function $a'_t(e, a)$ to get $\mu_t(e, \cdot)$ at each t

3. Aggregate policies $a'_t(e, a)$, $c_t(e, a)$ against $\mu_t(e, \cdot)$ at each t to get A_t , C_t

Key observation: r_t^p and Z_t determine everything

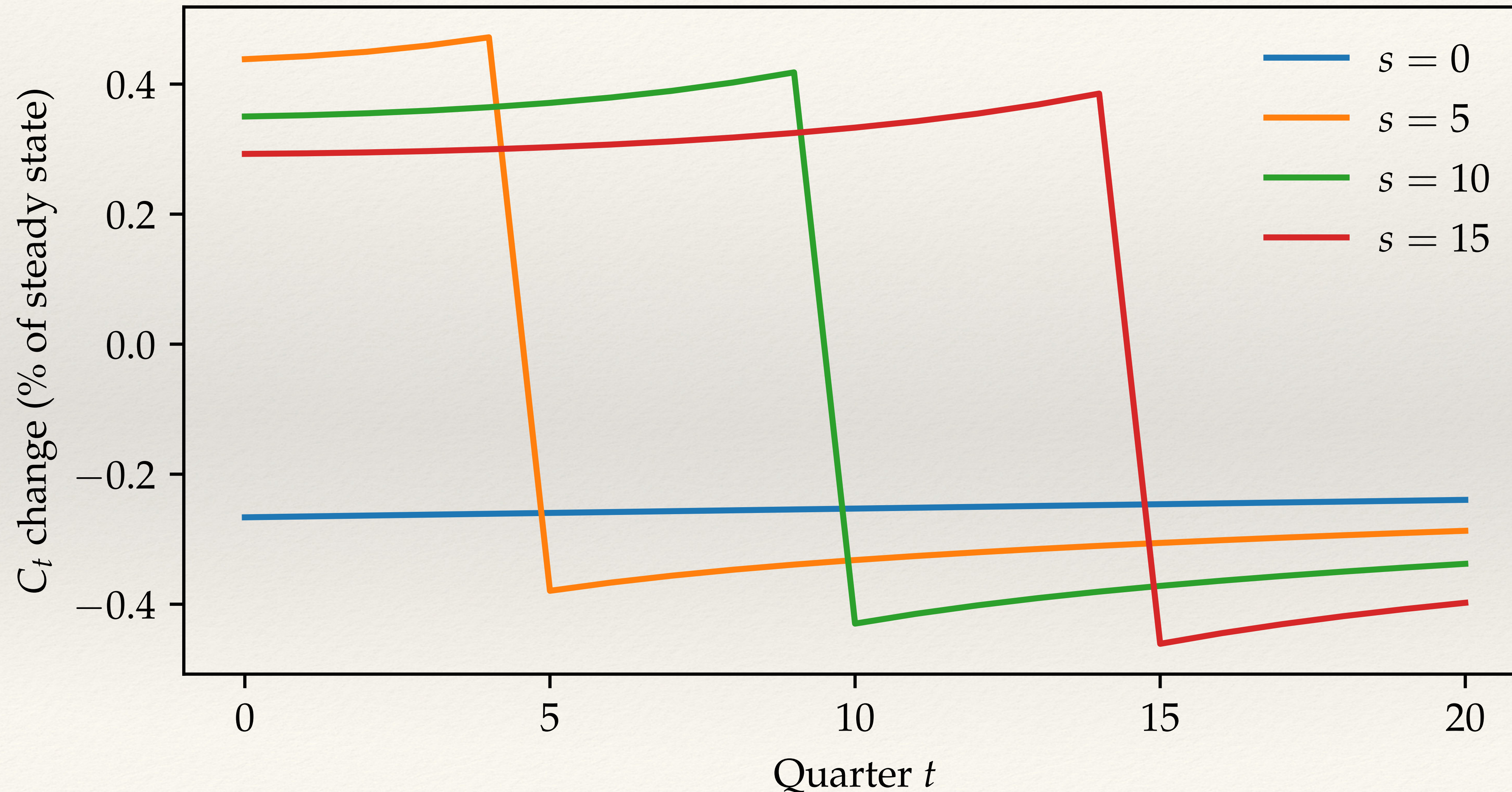
- ❖ Given sequences of ex-post returns r_t^p and aggregate after-tax income Z_t :
 - ❖ We can solve for time-varying policy functions $a'_t(e, a)$ and $c_t(e, a)$
 - ❖ These imply a time-varying distribution $\mu_t(e, \cdot)$
 - ❖ And together these imply time-varying A_t and C_t
- ❖ Hence, we can think of A_t and C_t as being **functions** of $\{r_s^p\}_{s=0}^{\infty}$ and $\{Z_s\}_{s=0}^{\infty}$
 - ❖ These are **sequence-space functions** $\mathcal{A}_t(\{r_s^p, Z_s\})$ and $\mathcal{C}_t(\{r_s^p, Z_s\})$
 - ❖ Plot around steady-state; later, compute derivatives (“sequence-space Jacobians”)

Response to 0.1% Z_s shocks at different dates s



Elevated spending
out of income at
other dates as well
(will call these
“intertemporal
MPCs”)

Response to -1 pp r_s^p shocks at different dates s



Consumption increases in anticipation of falling returns, but less than standard Euler equation

If $s = 0$ then it's a **surprise negative return** ("capital loss"). Implied MPC out of loss is only 0.011 , **very low** but persistent

Conclusion

Conclusion

- ❖ Introduced **standard incomplete markets model**
- ❖ Nice features: concave consumption functions, buffer-stock behavior, endogenous wealth distribution
- ❖ Resolve steady-state “asset-MPC tradeoff” by introducing β heterogeneity
- ❖ Aggregate dynamics a function of $\{r_s^p, Z_s\}$ path
- ❖ Elevated consumption in period of income shock, some before and after too
- ❖ Consumption response to r smaller than rep agent; MPC out of cap gains still low